

GEOGRAPHY

INTRODUCTION

Geography is about determining the significance of “place” as it relates to the natural environment, the human environment, and interactions within and between them. To investigate geographic issues, students must analyse the influences and interrelationships that give a place its distinctive characteristics and thus its spatial importance. Geographic analysis also requires an investigation of the economic, environmental, social, and political perspectives that relate to an issue. The application of the concepts of geographic thinking, spatial skills, and the use of field studies are central to the geographic inquiry process and the learning of geography.

Strands

The Grade 11 and 12 geography courses each have five strands. In all of these courses, strand A, Geographic Inquiry and Skill Development, is followed by four content strands, which are organized thematically.

Citizenship Education

The expectations in the Grade 11 and 12 geography courses provide opportunities for students to explore a number of concepts connected to the citizenship education framework (see page 13).

The Concepts of Geographic Thinking

The four concepts of geographic thinking – spatial significance, patterns and trends, interrelationships, and geographic perspective – underpin thinking and learning in all geography courses in the Canadian and world studies program. At least one concept of geographic thinking is identified as the focus for each overall expectation in the content strands of these courses. The following chart describes each concept and provides sample questions related to it. These questions highlight opportunities for students to apply a specific concept in their studies. (See page 16 for a fuller discussion of the concepts of disciplinary thinking.)

Spatial Significance
This concept requires students to determine the importance of a place or region. They explore the connections that exist between the geographical location and physical characteristics of a site and analyse the unique relationships that exist in and between the natural and human environments in a particular place. Students come to understand that the significance of the same place may be different for humans, animals, and plants. Related Questions* – What does a population settlement map tell us about the preferred range of latitude for settlement in Canada? (CGC1P, B2.2) – How does the official plan for your community address urban sprawl? (CGC1D, E2.3) – How might the uniqueness of a destination make it more appealing to some and less appealing to others? (CGG3O, B1.2) – How does the location of a site influence the feasibility of different types of power generation there? (CGR4E, E3.4) – What are some of the challenges that communities in permafrost regions face in providing safe drinking water and disposing of waste? (CGR4M, E2.3)
Patterns and Trends
This concept requires students to recognize characteristics that are similar and that repeat themselves in a natural or human environment (patterns) and characteristics or traits that exhibit a consistent tendency in a particular setting over a period of time (trends). The characteristics may be spatial, social, economic, physical, or environmental. Students analyse connections between characteristics to determine patterns; they analyse connections between those characteristics over time to determine trends. Related Questions – What trends do you see in the use of alternative energy in Ontario? (CGC1P, A1.5) – What pattern or patterns do you see in the location of First Nations reserves across Canada? (CGC1D, D3.1) – What factors have led to population movement in the region? What specific locations have attracted migrants? Why? (CGD3M, E2.1) – In what ways has technological change amplified the impact of population growth on the natural environment? In what ways has it diminished it? (CGW4U, C2.2)

* These questions are drawn directly from the overview charts that precede the geography courses and from the sample questions that accompany many specific expectations. To highlight the continuity between the geography courses in Grade 11 and 12 and those in Grade 9, and to show possible development in the use of the concepts of geographic thinking over those grades, the chart includes some questions from the Grade 9 geography curriculum as well.

Interrelationships

This concept requires students to explore connections within and between natural and human environments. The interconnected parts of an environment or environments work together to form a system. Students must understand the relationships that exist within a system and then critically analyse the relationships between systems in order to determine the impact they have on one another.

Related Questions

- How does surrounding farmland support a community, and what stresses might the community place on the farmland? (CGC1P, E2.1)
- How might the breaking up of continental ice in Greenland and the Antarctic affect Canada's coastline? (CGC1D, B2.1)
- Why do we need to understand the earth's physical processes in order to predict their potential hazards? (CGF3M, Overview)
- How can international trade policies and corporate actions influence disparities within or between ecumenes? (CGU4M, D1.2)

Geographic Perspective

This concept requires students to consider the environmental, economic, political, and/or social implications of the issues, events, developments, and/or phenomena that they are analysing. In order to solve problems, make decisions or judgements, or formulate plans of action effectively, students need to develop their ability to examine issues from multiple perspectives.

Related Questions

- In what ways can cultural diversity enrich the life of a community? (CGC1P, D1.2)
- Does the financial benefit of extracting natural resources justify related social and/or environmental impacts? (CGC1D, A1.6)
- How might spatial technologies help governments assess the economic, environmental, infrastructural, and social risks of a potential disaster? Why is it important for governments to know and understand these risks? (CGT3O, D1.2)
- Why might some people believe that there is an ethical imperative to share the world's resources equitably? (CGW4C, Overview)
- Why might military and/or government satellite information be shared with or withheld from certain countries or military alliances? What are the implications of doing so? (CGO4M, D2.1)

The Geographic Inquiry Process

In each of the geography courses in the Canadian and world studies curriculum, strand A focuses explicitly on the geographic inquiry process, guiding students in their investigations of issues, events, developments, and/or various geographic phenomena. This process is *not* intended to be applied in a linear manner: students will use the applicable components of the process in the order most appropriate for them and for the task at hand. Although strand A covers all of the components of the inquiry process, it is important to note that students apply skills associated with the inquiry process throughout the content strands in each course. (See page 32 for a fuller discussion of the inquiry process in the Canadian and world studies program.)

The following chart identifies ways in which students may approach each of the components of the geographic inquiry process.

Formulate Questions
Students formulate questions: – to explore various events, developments, issues, and/or phenomena that are related to the overall expectations in order to identify the focus of their inquiry – to help them determine which key concept or concepts of geographic thinking are relevant to their inquiry – that reflect the selected concept(s) of geographic thinking – to develop criteria that they will use in evaluating data, evidence, and/or information; in making judgements, decisions, or predictions; in reaching conclusions; and/or in solving problems
Gather and Organize
Students: – collect relevant qualitative and quantitative data, evidence, and information from field studies^a and a variety of primary and secondary sources,^b including visuals^c and community resources^d – determine if their sources are credible, accurate, and reliable – identify the purpose and intent of each source – identify the points of view in the sources they have gathered – use a variety of methods to organize the data, evidence, and/or information they have gathered – record the sources of the data, evidence, and information they are using – decide whether they have collected enough data, evidence, and/or information for their inquiry

a. Field studies may include, but are not limited to, studies in local neighbourhoods, school grounds, and various sites that allow students to explore different land uses (e.g., recreational, commercial, industrial, and transportation uses).

b. Primary sources may include, but are not limited to, census data, land claims, letters, photographs, speeches, and works of art. Secondary sources may include, but are not limited to, documentaries and other films, news articles, reference books, and most websites.

c. Visuals may include, but are not limited to, satellite images, maps, globes, models, graphs, and diagrams.

d. Community resources may include, but are not limited to, local conservation areas, resources from community groups and associations, government resources, and local plans.

Interpret and Analyse
Students: – analyse data, evidence, and information, applying the relevant concepts of geographic thinking (see preceding chart) – use different types of graphic organizers to help them interpret and/or analyse their data, evidence, and information – identify the key points or ideas in each source – analyse graphs, charts, diagrams, and maps – construct graphs, charts, diagrams, and maps to help them analyse the issue, event, development, or phenomenon they are investigating – analyse their sources to determine the importance of an issue, event, development, or phenomenon for individuals or groups – identify biases in individual sources – determine if all points of view are represented in the source materials as a whole, and which, if any, are missing
Evaluate and Draw Conclusions
Students: – synthesize data, evidence, and information, and make informed, critical judgements based on that data, evidence, and information – determine the short- and long-term impact of an event, development, issue, or phenomenon on people and/or places – reach conclusions about the subject of their inquiry, and support them with their data, evidence, and information – make predictions based on their data, evidence, and information – determine the ethical implications of an issue or action – determine the action required, where appropriate
Communicate
Students: – use appropriate forms (e.g., oral, visual, written, multimedia) for different audiences and purposes – communicate their arguments, conclusions, predictions, and plans of action clearly and logically – use geographical terminology and concepts correctly and effectively – cite sources, using appropriate forms of documentation

Forces of Nature: Physical Processes and Disasters, Grade 11

University/College Preparation

CGF3M

In this course, students will explore physical processes related to the earth's water, land, and air. They will investigate how these processes shape the planet's natural characteristics and affect human systems, how they are involved in the creation of natural disasters, and how they influence the impacts of human disasters. Throughout the course, students will apply the concepts of geographic thinking and the geographic inquiry process and use spatial technologies to analyse these processes, make predictions related to natural disasters, and assess ways of responding to them.

Prerequisite: Issues in Canadian Geography, Grade 9, Academic or Applied

OVERVIEW

The course has five strands. Instruction and learning related to the expectations in strand A are to be interwoven with instruction and learning related to expectations from the other four strands. Strand A must not be seen as independent of the other strands. Student achievement of the expectations in strand A is to be assessed and evaluated *throughout* the course.

Strand A

A. Geographic Inquiry and Skill Development
Overall Expectations
A1. Geographic Inquiry: use the geographic inquiry process and the concepts of geographic thinking when investigating physical systems, natural phenomena, and natural events
A2. Developing Transferable Skills: apply in everyday contexts skills, including spatial skills, developed through geographical investigation, and identify some careers in which a background in geography might be an asset

(continued)

Overview *(continued)*

Throughout this course, when planning instruction, teachers should weave the expectations from strand A in with the expectations from strands B–E.

Strands B–E

Overall Expectations and Related Concepts of Geographic Thinking	Big Ideas*	Framing Questions*
B. Spatial Organization: Spheres of the Earth		
B1. Physical Processes and Natural Hazards: analyse the characteristics of different types of natural hazards, and explain the role of physical processes in their occurrence (FOCUS ON: <i>Spatial Significance; Interrelationships</i>)	Natural hazards are a result of the earth's natural processes and phenomena.	Why do the earth's physical features change? How have various natural processes altered the earth's landscapes? What was created? What was destroyed?
B2. Spatial Connections: analyse relationships between physical processes and the earth's physical characteristics (FOCUS ON: <i>Patterns and Trends; Interrelationships</i>)	The earth's physical characteristics have been created and are changed by physical processes.	Why do we need to understand the earth's physical processes in order to predict their potential hazards?
B3. Physical Characteristics of the Earth: describe the spatial distribution of the earth's physical features and the processes that form them (FOCUS ON: <i>Spatial Significance; Interrelationships</i>)	Powerful natural processes formed the earth and continue to change and mould the outer spheres of the earth.	
C. The Physical Environment: Sustainability and Stewardship		
C1. Renewing the Physical Environment: analyse the role of physical processes and human practices in maintaining a sustainable natural environment (FOCUS ON: <i>Interrelationships; Geographic Perspective</i>)	The choices and actions of individuals, companies, and governments play a role in sustainability.	Why do humans modify the natural environment? Do we have a responsibility to protect the earth?
C2. Human Impact on the Physical Environment: analyse the impacts of human activities on the earth's physical processes and the natural environment (FOCUS ON: <i>Spatial Significance; Interrelationships</i>)	Human activity has an impact on the natural environment.	Why is it necessary for countries to work together on strategies to address sustainability?
C3. Human Use of the Physical Environment: analyse the influence of physical processes and features on human activity (FOCUS ON: <i>Spatial Significance; Geographic Perspective</i>)	The natural environment has an impact on human activity.	How do personal beliefs and values affect the choices that an individual or a government makes about resource use and sustainability?

* See page 17 for a discussion of the purpose of big ideas and framing questions.

Overall Expectations and Related Concepts of Geographic Thinking	Big Ideas*	Framing Questions*
D. Systems: Interaction and Interdependence		
D1. Sharing the Physical Environment: analyse issues relating to natural and human impacts on the environment and the sharing of natural resources between population groups (FOCUS ON: <i>Interrelationships; Geographic Perspective</i>)	What happens in one region can have a direct impact on another region.	Why do humans choose to live in disaster-prone areas? What are the potential impacts of human settlement in a disaster-prone area?
D2. Population and Disasters: analyse the influence of human settlement choices and the earth's physical processes on the impacts of disasters (FOCUS ON: <i>Patterns and Trends; Interrelationships</i>)	People in different regions of the world need different strategies to deal with potential disasters.	Who owns water? How do we control the use of freshwater resources?
D3. Earth's Planetary Characteristics and Life: explain the significance of Earth's planetary characteristics and history for the development and maintenance of life on Earth (FOCUS ON: <i>Spatial Significance; Interrelationships</i>)	We rely on the earth's physical characteristics and processes to sustain life.	What criteria would you use to rank potential geopolitical issues related to trends in human activity or changes in the environment?
E. Impacts of Change		
E1. Impacts of Processes and Disasters: analyse impacts of physical processes and disasters on human and natural systems, locally, nationally, and globally (FOCUS ON: <i>Interrelationships; Geographic Perspective</i>)	Natural processes and disasters can cause social, economic, political, and environmental problems.	Which potential impacts of climate change may affect your community or region?
E2. Disaster Preparedness: assess the role and effectiveness of various options for reducing the impacts of disasters on human populations (FOCUS ON: <i>Spatial Significance; Geographic Perspective</i>)	Disaster preparedness is everyone's responsibility.	How does studying disasters of the past help us prepare for disasters of the future?
E3. Processes of Change: describe how the earth's natural systems change, and have changed, over various time scales, and explain some of the processes that cause these changes (FOCUS ON: <i>Spatial Significance; Patterns and Trends</i>)	The earth today is a result of many natural processes. These processes continue to change the planet.	How do individuals and governments prepare for potential disasters? How have the earth's natural processes shaped the world we live in today?

A. GEOGRAPHIC INQUIRY AND SKILL DEVELOPMENT

OVERALL EXPECTATIONS

Throughout this course, students will:

- A1. Geographic Inquiry:** use the geographic inquiry process and the concepts of geographic thinking when investigating physical systems, natural phenomena, and natural events;
- A2. Developing Transferable Skills:** apply in everyday contexts skills, including spatial skills, developed through geographical investigation, and identify some careers in which a background in geography might be an asset.

SPECIFIC EXPECTATIONS

A1. Geographic Inquiry

Throughout this course, students will:

A1.1 formulate different types of questions to guide investigations into physical systems, natural phenomena, and natural events (e.g., factual questions: *What does the Gaia hypothesis describe?*; comparative questions: *What are the similarities and differences between tornadoes and hurricanes?*; causal questions: *How might climate change have an effect on various coastal cities around the world?*)

A1.2 select and organize relevant data and information on geographic issues from a variety of primary and secondary sources (e.g., primary: *raw data from fieldwork, both quantitative and qualitative; photographs; satellite images*; secondary: *published statistics, newspapers, books, atlases, geographic magazines, websites, graphs, charts, digital and print maps*), ensuring that their sources represent a diverse range of perspectives

Sample questions: “How might you use shipwreck data to analyse the location of dangerous water currents? Where might you find this data and information?” “What type of data and information do you need to collect in order to determine the best location for mining diamonds?” “What types of maps and graphs will help you analyse landslide patterns?”

A1.3 assess the credibility of sources and information relevant to their investigations (e.g., *by considering how the data are constructed to support the author’s point of view, the possible bias of the author, the expertise of the author, the accuracy of the text and supporting data, the intended audience, the purpose of the messaging, the context in which the information was presented*)

Sample questions: “Whose point of view does this source represent? Is the source biased? Have you consulted other sources that represent other points of view? Which source is most credible and why?”

A1.4 interpret and analyse data and information relevant to their investigations, using various tools, strategies, and approaches appropriate for geographic inquiry (e.g., *interpret weather maps indicating barometric pressure systems to predict areas of active weather; analyse graphs and charts of global temperatures over time to identify past ice ages and predict future glaciation trends; use charts to determine the costs and benefits of building structures that are designed to minimize the impact of natural disasters; use graphic organizers to outline various points of view on whether to send relief funds to a country where a natural disaster has occurred*)

Sample questions: “How might remote sensing help you analyse changes in water temperature along a shoreline?” “What data layers might you use with a geographic information system (GIS) in order to analyse impacts on an area affected by a wildfire?”

A1.5 use the concepts of geographic thinking (i.e., spatial significance, patterns and trends, interrelationships, geographic perspective) when analysing and evaluating data and information and formulating conclusions and/or judgements about physical systems, natural phenomena, and natural events (e.g., *use the concept of spatial significance to analyse the distribution of cold and hot deserts around the world; use the concept of patterns and trends to predict which regions of the earth will be most prone to future catastrophic seismic events; use the concept of interrelationships to analyse the impacts of fossil-fuel combustion on the earth's climate; use the concept of geographic perspective to analyse the social, political, economic, and environmental impacts of repeated flooding in an area*)

Sample questions: “How might the concept of spatial significance help you identify the boundaries of a drainage basin?” “How might an understanding of patterns and trends help you determine the best location for a house along a river?” “How might an understanding of interrelationships help you determine locations where solifluction might be an issue?” “How can geographic perspective help you analyse the impacts of climate change?”

A1.6 evaluate and synthesize their findings to formulate conclusions and/or make informed judgements or predictions about the issues they are investigating

Sample questions: “How might a knowledge of international agreements such as the United Nations Convention on the Law of the Sea or policies such as Canada’s Arctic sovereignty initiative help you in making judgements about some of the impacts of melting sea ice? Why might melting sea ice be an issue for some countries and not others? Which countries would you expect to be most concerned about the melting of the polar ice caps?”

A1.7 communicate their ideas, arguments, and conclusions using various formats and styles, as appropriate for the audience and purpose (e.g., *a debate with classmates on the need for natural spaces within their community; a report for the construction industry on the effects of acid precipitation and chemical weathering on various building materials; a webcast or podcast for the general public on ways of protecting themselves from a specific type of natural disaster; a photo essay for a local community group to illustrate the impacts of physical processes at work within their community*)

Sample questions: “What kind of information does your audience need? In how much detail? What format and approach would be most

effective in conveying your information to this particular audience?”

A1.8 use accepted forms of documentation (e.g., *footnotes, author/date citations, reference lists, bibliographies, annotated bibliographies, credits*) to reference different types of sources (e.g., *websites, blogs, books, articles, films, data*)

A1.9 use appropriate terminology when communicating the results of their investigations (e.g., *vocabulary specific to their inquiry; terminology related to geography and to the concepts of geographic thinking*)

A2. Developing Transferable Skills

Throughout this course, students will:

A2.1 describe ways in which geographic investigation can help them develop skills, including spatial skills and the essential skills in the Ontario Skills Passport (e.g., *reading graphic texts, writing, graphing, computer use, use of GIS, use of satellite imagery, oral communication, numeracy, decision making, planning, management, finding information, problem solving*), that can be transferred to postsecondary opportunities, the world of work, and everyday life

A2.2 apply in everyday contexts skills and work habits developed through geographic investigation (e.g., *use critical thinking, mapping, and graphing skills to analyse statistics and data in order to deepen their understanding of a local environmental issue; use listening skills to consider multiple perspectives when discussing a disaster caused by human error that affects the physical environment; apply work habits such as collaboration when working with a team to determine the criteria that need to be considered when making a decision relating to outdoor safety; use habits of self-regulation to monitor their own progress on a task*)

A2.3 apply the concepts of geographic thinking when analysing current events involving geographic issues (e.g., *use the concept of spatial significance to analyse the impact of pollutants released into the air; use the concept of patterns and trends to analyse potential impacts of El Niño and La Niña on North and South America; use the concept of interrelationships to analyse the connection between dropping water tables and sinking cities; use the concept of geographic perspective to analyse the potential consequences of harvesting a group of trees in the neighbourhood*) in order to enhance their understanding of these issues and their role as informed citizens

A2.4 identify some careers in which a geography background might be an asset (*e.g., conservation officer, land surveyor, landscaper, emergency preparedness coordinator, urban planner, civil engineer, insurance adjuster, oceanographer, seismologist, meteorologist, volcanologist*)

B. SPATIAL ORGANIZATION: SPHERES OF THE EARTH

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1. Physical Processes and Natural Hazards:** analyse the characteristics of different types of natural hazards, and explain the role of physical processes in their occurrence (**FOCUS ON:** *Spatial Significance; Interrelationships*)
- B2. Spatial Connections:** analyse relationships between physical processes and the earth's physical characteristics (**FOCUS ON:** *Patterns and Trends; Interrelationships*)
- B3. Physical Characteristics of the Earth:** describe the spatial distribution of the earth's physical features and the processes that form them (**FOCUS ON:** *Spatial Significance; Interrelationships*)

SPECIFIC EXPECTATIONS

B1. Physical Processes and Natural Hazards

FOCUS ON: *Spatial Significance; Interrelationships*

By the end of this course, students will:

- B1.1** analyse the characteristics of different types of weather and climatic hazards (*e.g., hurricanes, tornadoes, blizzards, droughts*), and explain the role of atmospheric conditions and processes in their occurrence (*e.g., temperature, pressure, and humidity contrasts; heat transfer through convection and condensation; sea surface temperatures; blocking; wind shear*)

Sample questions: "Why do Atlantic hurricanes tend to follow a boomerang-shaped path? Why do hurricanes lose strength after they move inland or cross over colder waters?" "Why does Oklahoma have more tornadoes than Nova Scotia?" "What hazards are associated with thunderstorms?" "Should forest fires be considered a climatic hazard?" "What is the relationship between tropical storms and hurricanes, storm intensity and atmospheric pressure, or cumulonimbus clouds and thunderstorm cells?"

- B1.2** analyse the characteristics of different types of geological hazards (*e.g., volcanoes, earthquakes, tsunamis, landslides, avalanches, mudflows*), and explain the role of tectonic and other geological processes in their occurrence (*e.g., folding, faulting, volcanism, mass wastage*)

Sample questions: "Why are some regions of the world more prone to volcanoes and earthquakes than others?" "What are some of the hazards associated with volcanoes?" "What are some of the common physical effects that accompany earthquakes, and what is their impact on human and natural systems in the vicinity of the earthquake?"

Using spatial skills: Students can add a map layer of regions prone to earthquakes and volcanoes to a base map of the earth's fault lines to investigate factors that affect the location of these events. Students can also identify the probable location of an underwater earthquake that triggered a tsunami by adding a map layer of regions struck by the tsunami to a base map of the earth's fault lines.

- B1.3** analyse the characteristics of different types of hydrological hazards (*e.g., storm surges, avalanches, flooding*), and explain the role of geological and/or atmospheric processes in their occurrence

Sample questions: "What are some of the ways in which flooding can occur?" "Why is the southern part of the Red River prone to flooding in the spring?" "Why is a snow avalanche both a hydrological and a geological hazard?" "How is climate change increasing the potential for damage by storm surges?"

B2. Spatial Connections

FOCUS ON: *Patterns and Trends; Interrelationships*

By the end of this course, students will:

B2.1 analyse the relationship between atmospheric processes and dynamics and the spatial distribution of major atmospheric and climatic phenomena (e.g., *trade winds, jet streams, and Hadley cells; low-pressure systems and different types of storms; cooling and glaciation*)

Sample questions: “How does the differential heating and cooling of the earth’s surface explain the global wind belts?” “How does albedo contribute to the differences in climate between polar and equatorial regions?”

B2.2 analyse the relationship between the earth’s geological processes (e.g., *rock and mineral formation, plate tectonics, gradational processes*) and the spatial distribution of major physical features and phenomena (e.g., *features: mountain ranges and the Pacific Ring of Fire, river systems, volcanic mountains; phenomena: earthquakes, volcanic eruptions, landslides*)

Sample questions: “What kinds of maps and spatial technologies would be most useful for determining the relationship between the pattern of earthquake and volcanic occurrences and crustal plate boundaries?” “Why are there different types of earthquakes and volcanoes?” “How might monitoring and investigating fault lines in Iceland and California help in understanding processes inside the earth?” “How do the geological characteristics of an area determine river drainage patterns?”

Using spatial skills: Students can analyse topographic maps of areas where glacial deposition has occurred in order to understand the connection between glaciation and surface features such as kames, moraines, eskers, and erratics. Students can also use topographic maps to support their investigation of the role of erosion in various stages of river formation.

B2.3 analyse the relationship between the earth’s hydrospheric processes and the spatial distribution of related major physical features and phenomena (e.g., *the relationship between ocean currents and coastal characteristics, ocean currents and climatic patterns, gravitational pull and tides*)

Sample questions: “Why could glaciation be considered an atmospheric as well as a hydrologic process?” “How are tide heights related to the phases of the moon?” “How do rivers alter landforms?”

B2.4 analyse the effects of biological processes on different components of the Earth system (e.g., *the role of trees, decaying plants, mosses, lichens, microorganisms, and worms in weathering and soil formation; the role of animals in the modification of surface features; the influence of living organisms on the composition of the atmosphere and ocean chemistry*)

B3. Physical Characteristics of the Earth

FOCUS ON: *Spatial Significance; Interrelationships*

By the end of this course, students will:

B3.1 describe the regional distribution of the earth’s major physical characteristics (e.g., *landforms, water bodies and systems, climate regions, vegetation regions*), and analyse factors that influence their spatial distribution

Sample questions: “What natural processes played a role in the formation of the Sahara? Where else have similar forces played a role in the formation of a desert?” “In which regions of the world would you expect orographic precipitation to occur?” “Using maps of global wind patterns, can you identify countries that are normally affected by monsoon rains?” “What natural factors determine the spatial extent of a rainforest?”

Using spatial skills: Students can identify spatial patterns relating to physical features by overlaying a base map of the earth with maps of different physical features. They can identify further relationships associated with a specific feature by overlaying the map of that feature with maps of other features that might be related to it.

B3.2 describe the characteristics of the earth’s outer spheres (e.g., *lithosphere, hydrosphere, atmosphere, biosphere*) and their relationship to the earth’s internal spheres (i.e., *core, mantle*), and explain the role of both in the early phases of the earth’s development

Sample questions: “What processes in the earth’s interior led to the formation of the continents?” “How did processes in the earth’s interior contribute to the development of our present atmosphere?” “What conditions were necessary for the development of living organisms?” “How did early life forms modify the atmosphere?”

C. THE PHYSICAL ENVIRONMENT: SUSTAINABILITY AND STEWARDSHIP

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1. Renewing the Physical Environment:** analyse the role of physical processes and human practices in maintaining a sustainable natural environment (**FOCUS ON:** *Interrelationships; Geographic Perspective*)
- C2. Human Impact on the Physical Environment:** analyse the impacts of human activities on the earth's physical processes and the natural environment (**FOCUS ON:** *Spatial Significance; Interrelationships*)
- C3. Human Use of the Physical Environment:** analyse the influence of physical processes and features on human activity (**FOCUS ON:** *Spatial Significance; Geographic Perspective*)

SPECIFIC EXPECTATIONS

C1. Renewing the Physical Environment

FOCUS ON: *Interrelationships; Geographic Perspective*

By the end of this course, students will:

- C1.1** describe ways in which the physical environment renews itself naturally (*e.g., decaying matter develops into soil; silt from erosional forces may form rock; biogeochemical cycles such as the carbon, nitrogen, and hydrologic cycles transfer water and nutrients back and forth between the living and non-living components of the environment*)

Sample questions: “How has life on Earth continued for billions of years when the resources needed to support it are virtually finite?” “If mineral resources can be renewed naturally, why do we categorize them as a non-renewable resource?” “How do forest fires contribute to forest regeneration?” “In what ways is destruction and renewal a part of the evolutionary adaptation of natural systems?”

- C1.2** assess the effectiveness of various processes used by resource-based industries to protect or rehabilitate the physical environment

Sample questions: “What are some of the environmental problems associated with the closing of open-pit mines, and what options are available for rehabilitating these sites in order to minimize long-term environmental damage?” “Is ‘green’ mining possible? How can the environmental impacts of mining be reduced?” “Can we make our existing reserves of non-renewable resources last longer? In what ways would that benefit the environment?” “How does contour farming protect against soil

erosion?” “What criteria would you use to assess the effectiveness of a rehabilitation process?”

- C1.3** identify and assess the effectiveness of various national and international strategies and initiatives for reducing the human impact on the natural environment (*e.g., protected areas, such as national parks, marine protection zones, and UNESCO biosphere reserves; environmental monitoring programs; national legislation such as the Species at Risk Act; international agreements such as the Canada–United States Air Quality Agreement or the Great Lakes Water Quality Agreement*)

Sample questions: “Should the international community support the trading of carbon credits?” “Why did the Kyoto Protocol and the Montreal Protocol have different outcomes?”

- C1.4** describe strategies and practices that are used to protect local ecosystems and make them more sustainable (*e.g., watershed and wildlife management programs; conservation areas; wildlife corridors; sustainable agricultural practices, such as no-till farming and use of buffer strips to separate fields from rivers; urban strategies such as municipal transit subsidies, waste-reduction programs, and pesticide bans*)

Sample question: “How do green belts and the curbing of urban sprawl contribute to the sustainability of natural systems in heavily urbanized regions?”

- C1.5** identify strategies and practices that individuals can use to reduce their personal impact on the natural environment (*e.g., changing personal*

consumption patterns, minimizing waste through reuse and recycling, supporting environmental advocacy organizations)

Sample questions: “What are some ways in which individuals can work to reduce emissions of greenhouse gases and smog-causing pollutants?” “What are some simple things that you can do to reduce water pollution?” “What common household items contain toxic materials? How should they be disposed of? Are there alternatives to these items that are not toxic or don’t require hazardous waste disposal?” “What is urban farming and what financial benefit does it have for families? How does urban farming help reduce environmental waste?”

C2. Human Impact on the Physical Environment

FOCUS ON: *Spatial Significance; Interrelationships*

By the end of this course, students will:

C2.1 explain how human modification of the landscape (e.g., urban expansion, paving, river damming or diversion, deforestation, draining of wetlands, mining) affects natural systems (e.g., changes in groundwater, loss of habitat, alteration of food chains, creation of heat domes and other local climatic changes, changes in carbon budget and related climate effects, changes in drainage patterns, impairment of surface water quality)

Sample questions: “In what ways did the modification of the Yangtze River, due to the development of the Three Gorges Dam, affect natural systems in that region of China? What criteria would you use to determine which of these changes will have the largest and longest-lasting impacts?”

C2.2 explain how human activities that modify the natural environment may increase the possibility of a natural disaster or intensify its impacts (e.g., deforestation or urban construction that alters bedrock may increase the possibility of a landslide; building houses on permafrost may trigger mudflows; draining and infilling of wetlands may increase the severity of floods)

Sample questions: “How did coal mining in Turtle Mountain, Alberta, contribute to the Frank Slide?” “How did the Vajont Dam in Italy contribute to a flooding disaster, and why was the site of the dam a poor choice?” “How did the construction of offshore islands as part of a land-reclamation process contribute to liquefaction impacts during the Great Hanshin Earthquake of 1995?”

C2.3 assess the effectiveness of various means for decreasing risks from natural hazards (e.g., bans on housing construction in potentially dangerous areas such as floodplains or beachfronts; building code provisions; protective infrastructure such as breakwaters; watershed management practices such as flood control, flood forecasting, and erosion control; avalanche and landslide control measures)

Sample questions: “What impact did the building of a tsunami wall have on the Japanese island community of Okushiri?” “Why did levees fail to protect New Orleans during Hurricane Katrina?”

C3. Human Use of the Physical Environment

FOCUS ON: *Spatial Significance; Geographic Perspective*

By the end of this course, students will:

C3.1 describe how various features in the physical environment provide economic resources or benefits for society (e.g., waterfalls provide energy sources; moraines and eskers provide sand and gravel; forests and minerals provide materials for building and manufacturing; sheltered bays provide good harbours)

Sample questions: “What are the economic benefits of a river?” “What kinds of physical features attract human settlement? What are some of the economic benefits associated with these features?” “What are the economic advantages of your community’s physical environment?”

C3.2 analyse the costs and benefits of various natural phenomena from a human perspective (e.g., volcanic eruptions pose a danger to human settlement but produce fertile soil; monsoon rains lead to flooding of the land but are vital to crop production; subduction at tectonic plate boundaries causes earthquakes and volcanoes but also forms concentrations of minerals)

Sample questions: “Can excessive snowfall provide benefits to humans? Are there costs related to below-average snowfall?”

D. SYSTEMS: INTERACTION AND INTERDEPENDENCE

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1. Sharing the Physical Environment:** analyse issues relating to natural and human impacts on the environment and the sharing of natural resources between population groups (**FOCUS ON:** *Interrelationships; Geographic Perspective*)
- D2. Population and Disasters:** analyse the influence of human settlement choices and the earth's physical processes on the impacts of disasters (**FOCUS ON:** *Patterns and Trends; Interrelationships*)
- D3. Earth's Planetary Characteristics and Life:** explain the significance of Earth's planetary characteristics and history for the development and maintenance of life on Earth (**FOCUS ON:** *Spatial Significance; Interrelationships*)

SPECIFIC EXPECTATIONS

D1. Sharing the Physical Environment

FOCUS ON: *Interrelationships; Geographic Perspective*

By the end of this course, students will:

- D1.1** explain how natural events or human activities in one part of the world can have impacts on other parts of the world (*e.g., downstream impacts of dams, climatic effects of volcanic eruptions, effects of fossil-fuel combustion on global climate and acidity of precipitation, effects of the use of ozone-depleting chemicals on the ozone layer*)

Sample questions: "What impact does a large volcanic eruption have on climate, and what natural processes are responsible for this impact?" "What impact is rapid economic development in Brazil, Russia, India, and China having on global climate change?"

- D1.2** analyse issues relating to the sharing of resources by different countries or population groups (*e.g., water extraction, water level, and water quality issues in the Great Lakes basin; resource exploration and extraction and wildlife management in the Arctic; conflict between herders and farmers in the Sahel region of Africa; conflict between ranchers and ecologists over the reintroduction of wolves to Yellowstone National Park*)

Sample questions: "Who owns the oceans?" "Should governments allow resource extraction

in sensitive environments such as the High Arctic or Antarctica?" "How might Aboriginal treaty rights affect agreements between Canada and the United States for managing shared water resources?" "How have various countries tried to manage shared water resources such as the Rhine, Brahmaputra, and Jordan rivers?" "What were the main issues surrounding the Cochabamba Water War in Bolivia? How did this conflict contribute to the UN's recognition in 2010 that water is a human right?"

- D1.3** make predictions about geopolitical issues that may emerge in the future and the regions that they will affect on the basis of current trends in human activities and environmental conditions

Sample questions: "How will climate change affect the availability of water in different parts of the world, and what issues will countries in water-rich and water-poor regions face as a result?" "What countries are likely to face severe challenges due to rising sea levels, and what are their options for responding?" "How does the distribution of wealth affect the amount and quality of water that different population groups have access to, and how might it affect their ability to respond to related challenges due to climate change?"

D2. Population and Disasters

FOCUS ON: *Patterns and Trends; Interrelationships*

By the end of this course, students will:

D2.1 analyse factors that influence human settlement in hazardous areas (*e.g., slopes of active volcanoes, floodplains, Tornado Alley, coastal areas of Bangladesh*), and assess ways of moderating the risks of living in these areas

Sample questions: “Why might people live in an area that they know is prone to natural disasters?” “What is the difference between a natural hazard and a natural disaster?” “How can people adapt to natural hazards in order to reduce the risk of disasters? What determines whether they can adapt effectively or not? Should people be allowed to settle in areas where there is a high risk of a natural disaster?”

Using spatial skills: Using a world map as a base, students can identify disaster-prone regions by adding layers showing where different types of natural disasters have occurred. They can also use this information to identify the place where, in their opinion, people face the highest natural disaster risk in the world.

D2.2 explain how the earth’s physical processes can affect the impacts of a human disaster (*e.g., wave action and BP’s Gulf of Mexico oil spill of 2010; wind patterns and Japan’s Fukushima nuclear disaster of 2012; wind, precipitation, and forest fire spread*)

Sample question: “Why is it important to understand the earth’s physical processes when dealing with disasters caused by humans?”

Using spatial skills: On a map of the affected region, students can plot the point of origin of a disaster and shade in the areas that experienced impacts. The map can be used to support further investigations of the processes that determined which areas were affected and in what ways.

D3. Earth’s Planetary Characteristics and Life

FOCUS ON: *Spatial Significance; Interrelationships*

By the end of this course, students will:

D3.1 explain the biological significance of Earth’s position in space (*e.g., with respect to temperature, presence of liquid water, presence of a life-supporting atmosphere*)

Sample questions: “Why is the story of Goldilocks often referred to when describing Earth’s position in space?” “What are the characteristics of Earth that would need to change in order for Earth to be like Venus or Mars?”

D3.2 explain how the earth’s shape, orbit, rotation, and tilt produce seasonal and regional variations in climate and support biodiversity by creating regionally diverse conditions for life

Sample questions: “What is the significance of the following lines of latitude to the seasons – the tropic of Cancer, the tropic of Capricorn, the equator, the Arctic and Antarctic circles?” “How does the tilt of the earth’s axis cause seasons?” “Why is it summer in the Northern Hemisphere when it is winter in the Southern Hemisphere?” “Which parts of the earth have the greatest quantity and diversity of living organisms? Why?”

D3.3 demonstrate a critical understanding of various theories of the origins of the universe and of Earth’s formation and evolution (*e.g., formation: big bang, nebular hypothesis; evolution: continental drift, plate tectonics*), and describe the theories’ contribution to our understanding of some of Earth’s major characteristics

Sample questions: “What are the strengths and weaknesses of the evidence used to support the theory of plate tectonics?” “How has continental drift contributed to biological diversity?” “How does the nebular hypothesis explain the varying densities of the earth’s layers?”

E. IMPACTS OF CHANGE

OVERALL EXPECTATIONS

By the end of this course, students will:

- E1. Impacts of Processes and Disasters:** analyse impacts of physical processes and disasters on human and natural systems, locally, nationally, and globally (**FOCUS ON:** *Interrelationships; Geographic Perspective*)
- E2. Disaster Preparedness:** assess the role and effectiveness of various options for reducing the impacts of disasters on human populations (**FOCUS ON:** *Spatial Significance; Geographic Perspective*)
- E3. Processes of Change:** describe how the earth's natural systems change, and have changed, over various time scales, and explain some of the processes that cause these changes (**FOCUS ON:** *Spatial Significance; Patterns and Trends*)

SPECIFIC EXPECTATIONS

E1. Impacts of Processes and Disasters

FOCUS ON: *Interrelationships; Geographic Perspective*

By the end of this course, students will:

- E1.1** analyse selected short- and long-term impacts of natural processes in various parts of the world (*e.g., river erosion and the development of canyons; rising sea level and the loss of islands; volcanic activity and the loss of islands, the building of islands, and increases in soil capability; glaciation and till deposition and soil formation; earthquakes and separation of land; erosion and sedimentation and the development of deltas, floodplains, and other landforms; sedimentation and fossil-fuel formation; plate tectonics and mineral formation*)

Sample questions: "What are some natural processes that affect your local area? What impacts might these have on your local community or region over time?"

- E1.2** compare, from a geographic perspective, the impacts of selected natural and human disasters (*e.g., Hurricane Sandy and Hurricane Katrina; earthquakes in Haiti in 2010 and in Kobe, Japan, in 1995; recent droughts in the Sahel and the dust bowl in North America during the 1930s; the 2010 BP oil spill in the Gulf of Mexico and the grounding of the Exxon Valdez in 1989*)

Sample questions: "What were the impacts of the Indian Ocean earthquake and tsunami of 2004? How did it affect the people, their homes, their schools, and businesses? What political impact did this disaster have on the region in which it occurred? Was the economic impact felt only in the region or did it have global

implications? In what ways did the damage caused by this disaster affect the natural environment? How do these impacts compare with those of the Japan earthquake and tsunami of 2011?"

- E1.3** assess potential impacts, both positive and negative, of climate change on different parts of the world, including their local community

Sample questions: "How will Arctic warming affect animal populations and migration patterns? How are indigenous communities likely to be affected? What are the consequences of thawing permafrost? What are some of the economic benefits that are expected to come from the warming of the Arctic?" "What areas of the world currently produce the most grain foods? How might this change? What concerns might a change in grain production raise?" "How might climate change have an impact on tourist destinations that depend on the weather?" "Why are some coastal cities more vulnerable to the rise of sea level than others? Why are poorer regions more vulnerable than richer ones?" "Which countries might be faced with environmental migration issues due to climate change?" "How are regions that depend on glacier-fed rivers for their water supply likely to be affected by climate change?"

Using spatial skills: Students can make a preliminary identification of coastal settlements that are vulnerable to flooding from rising sea levels by adding a map showing coastal elevation data to a world base map showing human settlement.

E2. Disaster Preparedness

FOCUS ON: *Spatial Significance; Geographic Perspective*

By the end of this course, students will:

E2.1 explain how various technologies, devices, and classification systems (e.g., *tilt meters, laser rangefinders, seismographs, the global positioning system [GPS], GIS, satellites, modified Richter scale, Saffir-Simpson scale, Fujita scale, anemometers, barometers, wet and dry thermometers*) are used to measure, monitor, predict, and analyse natural processes and hazards, and assess their role in reducing the human impacts of disasters

Sample questions: “What technology and data would you need to set up a tsunami warning system? Why might such a system be able to save lives?” “What role does Doppler radar play in forecasting storm activity?”

E2.2 assess the role of various adaptation and response measures in reducing the human impacts of natural disasters (e.g., *research, monitoring, communication, geo-engineering, levees, dams, building codes, windbreaks, cloud seeding, avalanche control*)

Sample questions: “How can governments use earthquake data to evaluate risks and develop appropriate emergency response measures?” “Why is it important for people to know the difference between a tornado watch and a tornado warning?”

E2.3 analyse ways in which communities prepare for and respond to natural disasters, using local, national, and international examples

Sample questions: “Does your community have an emergency plan to respond to natural disasters? Which hazards does it specifically address? How effective do you think the plan would be in the event of a real disaster? Are there potential hazards it does not address?” “How do your community’s plans for responding to a particular type of disaster compare with those of other communities in Canada or in other parts of the world that face similar risks? Do other communities have plans or systems in place that should be considered by your community?”

E3. Processes of Change

FOCUS ON: *Spatial Significance; Patterns and Trends*

By the end of this course, students will:

E3.1 analyse patterns and trends in the earth’s natural systems over geologic time

Sample questions: “How have changes in the earth’s physical characteristics influenced biological change? How have biological changes influenced the earth’s physical processes?” “Climate change has directly or indirectly contributed to past mass extinctions in geologic history. Do you think that human-induced climate change will lead to an extinction in the future? Why or why not?” “How might human activity be shaping future changes in the earth’s natural systems?”

Using spatial skills: Students can reinforce their understanding of change over geologic time by using illustrations to mark significant geological and biological events on a representation of the geologic time scale. Students can also graph global climate data using different scale intervals in order to make inferences relating to trends in climate.

E3.2 explain how the earth’s tilt and changes in its position with respect to the sun and moon influence natural changes in the physical environment (e.g., *seasons, tide heights, ocean currents, the ozone layer*)

E3.3 explain the role of various erosional and depositional agents (e.g., *water, ice, wind, chemical processes*) in shaping the earth’s physical features (e.g., *meanders, valleys, dunes, caves*), locally, regionally, and globally

Sample questions: “How can glaciation be both an agent of degradation and erosion and an agent of deposition?” “What roles does water play in various physical processes?”